Docker Tasks - 03

- Create an image from a running container.

```
[ec2-user@ip-172-31-24-34 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS                                       NAMES
a685fbd30392   nginx     "/docker-entrypoint.…"   13 seconds ago   Up 12 seconds   0.0.0.0:8080->80/tcp, :::8080->80/tcp       nginx
[ec2-user@ip-172-31-24-34 ~]$ docker commit nginx myimage:v1
sha256:e024a6108ac78f84308f27a2e0cb4fed248ebb1791e3c4cb96558cec7d14c4c9
[ec2-user@ip-172-31-24-34 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS                                       NAMES
a685fbd30392   nginx     "/docker-entrypoint.…"   52 seconds ago   Up 50 seconds   0.0.0.0:8080->80/tcp, :::8080->80/tcp       nginx
[ec2-user@ip-172-31-24-34 ~]$ docker images
REPOSITORY                                                  TAG       IMAGE ID       CREATED          SIZE
myimage                                                     v1        e024a6108ac7   12 seconds ago   161MB
526018540742.dkr.ecr.us-east-1.amazonaws.com/mynginx        latest    64048fae9563   2 days ago       341MB
mynginx                                                     latest    64048fae9563   2 days ago       341MB
viqaastaimoor/mynginx                                       latest    64048fae9563   2 days ago       341MB
tomcat                                                      latest    d10cfd9141f2   5 days ago       417MB
jenkins/jenkins                                             latest    0a60439ca04e   8 days ago       491MB
sonarqube                                                   latest    92e8e49c2237   9 days ago       1.45GB
nginx                                                       latest    1a1e63136420   9 days ago       161MB
ubuntu                                                      latest    f794f40ddfff   4 weeks ago      78.1MB
```

- Copy image from local machine to docker server and load the image.

```
Viqaas@LAPTOP-VGO25G9U MINGW64 /d/DevOps/Keys
$ ls
Docker_Token.txt  Newkey.pem  Slave.pem  image.png   myimage.tar
Linux-key.pem     Peer.pem    Win.pem    my-key.pem

Viqaas@LAPTOP-VGO25G9U MINGW64 /d/DevOps/Keys
$ scp -i Linux-key.pem myimage.tar ec2-user@54.204.113.67:/home/ec2-user/
myimage.tar                                   100%  157MB   4.3MB/s   00:36
```

```
[ec2-user@ip-172-31-24-34 ~]$ ls
image.png  myimage.tar  sample.war
[ec2-user@ip-172-31-24-34 ~]$
```

```
[ec2-user@ip-172-31-24-34 ~]$ docker load -i myimage.tar
Loaded image: myimage:v1
[ec2-user@ip-172-31-24-34 ~]$ docker images
REPOSITORY                                                  TAG       IMAGE ID       CREATED          SIZE
myimage                                                     v1        e024a6108ac7   10 minutes ago   161MB
526018540742.dkr.ecr.us-east-1.amazonaws.com/mynginx        latest    64048fae9563   2 days ago       341MB
mynginx                                                     latest    64048fae9563   2 days ago       341MB
viqaastaimoor/mynginx                                       latest    64048fae9563   2 days ago       341MB
tomcat                                                      latest    d10cfd9141f2   5 days ago       417MB
jenkins/jenkins                                             latest    0a60439ca04e   8 days ago       491MB
sonarqube                                                   latest    92e8e49c2237   9 days ago       1.45GB
nginx                                                       latest    1a1e63136420   9 days ago       161MB
ubuntu                                                      latest    f794f40ddfff   4 weeks ago      78.1MB
```

- Create Docker image using alpine and customize with tomcat.

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```
[ec2-user@ip-172-31-24-34 ~]$ cat Dockerfile
FROM alpine

RUN apk add openjdk11 wget && \
    wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.80/bin/apache-tomcat-9.0.80.tar.gz && \
    tar -xzf apache-tomcat-9.0.80.tar.gz && \
    mv apache-tomcat-9.0.80 /tomcat

EXPOSE 8080

CMD ["/tomcat/bin/catalina.sh", "run"]
[ec2-user@ip-172-31-24-34 ~]$ docker build -t alpine .
[+] Building 72.9s (6/6) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 380B
 => [internal] load metadata for docker.io/library/alpine:latest
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/2] FROM docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
 => => resolve docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
 => => extracting sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153
 => => sha256:59855d3dceb3ae53991193bd03301e082b2a7faa56a514b03527ae0ec2ce3a95 1.02kB / 1.02kB
 => => sha256:a40c03cbb81c59bfb0e0887ab0b1859727075da7b9cc576a1cec2c771f38c5fb 611B / 611B
 => => sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153 3.86MB / 3.86MB
 => => sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659 9.22kB / 9.22kB
 => [2/2] RUN apk add openjdk11 wget &&     wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.80/bin/apache-tomcat-9.0.80.tar.gz &&
 => exporting to image
 => => exporting layers
 => => writing image sha256:f5ac3381debcafe4e9dd10f7ada1d0577757b438144dc9612e84de207af5b1ab
 => => naming to docker.io/library/alpine
```

- Create single stage and multistage docker file using this source code:

  https://github.com/betawins/multi-stage-example.git

- Single stage Dockerfile:

```
[ec2-user@ip-172-31-24-34 multi-stage-example]$ ls
Dockerfile  README.md  mvnw  mvnw.cmd  pom.xml  src
[ec2-user@ip-172-31-24-34 multi-stage-example]$ cat Dockerfile
FROM maven:3.8.6-openjdk-11

WORKDIR /app
COPY . .
RUN mvn clean package

CMD ["java", "-jar", "target/app.jar"]
[ec2-user@ip-172-31-24-34 multi-stage-example]$ docker build -t myapp .
[+] Building 62.5s (9/9) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 209B
 => [internal] load metadata for docker.io/library/maven:3.8.6-openjdk-11
 => [internal] load .dockerignore
 => => transferring context: 2B
```

- Multistage Dockerfile:

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```
[ec2-user@ip-172-31-24-34 multi-stage-example]$ cat Dockerfile
FROM maven:3.8.6-openjdk-11 AS build

WORKDIR /app
COPY . .
RUN mvn clean package

FROM eclipse-temurin:11-jre

COPY --from=build /app/target/*.jar app.jar

CMD ["java", "-jar", "app.jar"]
[ec2-user@ip-172-31-24-34 multi-stage-example]$ docker build -t app:v2 .
[+] Building 44.6s (12/12) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 287B
 => [internal] load metadata for docker.io/library/eclipse-temurin:11-jre
 => [internal] load metadata for docker.io/library/maven:3.8.6-openjdk-11
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [build 1/4] FROM docker.io/library/maven:3.8.6-openjdk-11@sha256:805f366910aea2a91ed263654
 => [stage-1 1/2] FROM docker.io/library/eclipse-temurin:11-jre@sha256:52c91c17a87050336a2f95e
```

- Install docker compose and execute sample application.

```
[ec2-user@ip-172-31-24-34 ~]$ mkdir -p ~/.docker/cli-plugins
[ec2-user@ip-172-31-24-34 ~]$ curl -SL https://github.com/docker/compose/releases/latest/download/docker-compose-linux-x86_64 \
-o ~/.docker/cli-plugins/docker-compose
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0     0     0  --:--:--  --:--:--  --:--:--     0
  0     0    0     0    0     0     0     0  --:--:--  --:--:--  --:--:--     0
100 30817k 100 30817k    0     0 76152k     0  --:--:--  --:--:--  --:--:-- 76152k
[ec2-user@ip-172-31-24-34 ~]$ chmod +x ~/.docker/cli-plugins/docker-compose
[ec2-user@ip-172-31-24-34 ~]$ docker compose version
Docker Compose version v5.1.1
```

```
[ec2-user@ip-172-31-24-34 app]$ cat docker-compose.yml
version: "3"
services:
  web:
    build: .
    ports:
      - "5000:5000"
[ec2-user@ip-172-31-24-34 app]$ docker compose up -d
WARN[0000] /home/ec2-user/app/docker-compose.yml: the attribute `version`
[+] Building 31.4s (11/11) FINISHED
 => [internal] load local bake definitions
 => => reading from stdin 468B
 => [internal] load build definition from Dockerfile
```

```
 => => naming to docker.io/library/app-web
 => resolving provenance for metadata file
[+] up 3/3
 ✔ Image app-web        Built
 ✔ Network app_default  Created
 ✔ Container app-web-1  Started
[ec2-user@ip-172-31-24-34 app]$
```

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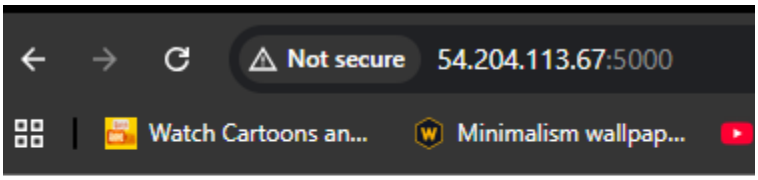

Hello from Docker!

```
[ec2-user@ip-172-31-24-34 app]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS                                       NAMES
6a248fd578d0   app-web   "python app.py"          27 seconds ago   Up 26 seconds   0.0.0.0:5000→5000/tcp, :::5000→5000/tcp     app-web-1
a685fbd30392   nginx     "/docker-entrypoint.…"   36 minutes ago   Up 36 minutes   0.0.0.0:8080→80/tcp, :::8080→80/tcp         nginx
[ec2-user@ip-172-31-24-34 app]$
```

- Implement solution to scan images when pushed to docker registry.

```
[ec2-user@ip-172-31-24-34 ~]$ docker images
\REPOSITORY   TAG       IMAGE ID       CREATED          SIZE
app-web       latest    c4fa32ca5f3f   14 minutes ago   1.1GB
app           v2        595b29f33006   24 minutes ago   283MB
nginx         latest    1a1e63136420   9 days ago       161MB
[ec2-user@ip-172-31-24-34 ~]$ docker scout cves app:v2
    ✔ Image stored for indexing
    ✔ Indexed 238 packages
    ✘ Detected 29 vulnerable packages with a total of 106 vulnerabilities


## Overview

                    |        Analyzed Image
--------------------|---------------------------------------------------
 Target             |  app:v2
   digest           |  595b29f33006
   platform         | linux/amd64
   vulnerabilities  |    7C    53H    58M    12L
   size             | 125 MB
   packages         | 238
```

```
130 vulnerabilities found in 29 packages
  CRITICAL   7
  HIGH       53
  MEDIUM     58
  LOW        12


What's next:
    View base image update recommendations → docker scout recommendations app:v2

[ec2-user@ip-172-31-24-34 ~]$
```

- Implement solution to scan images when pushed to AWS ECR.

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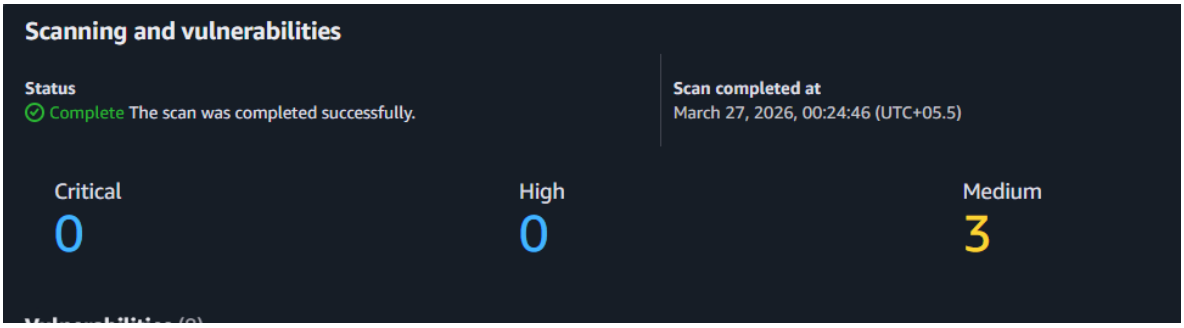
```
Login Succeeded
[ec2-user@ip-172-31-24-34 ~]$ docker tag app:v2 526018540742.dkr.ecr.us-east-1.amazonaws.com/myrepo:latest
[ec2-user@ip-172-31-24-34 ~]$ aws ecr put-image-scanning-configuration \
--repository-name myrepo \
--image-scanning-configuration scanOnPush=true
{
    "registryId": "526018540742",
    "repositoryName": "myrepo",
    "imageScanningConfiguration": {
        "scanOnPush": true
    }
}
[ec2-user@ip-172-31-24-34 ~]$ docker push 526018540742.dkr.ecr.us-east-1.amazonaws.com/myrepo:latest
The push refers to repository [526018540742.dkr.ecr.us-east-1.amazonaws.com/myrepo]
17961c817f2b: Pushed
61a6f8affb50: Pushed
56437b53119b: Pushed
ec6e2d3734df: Pushed
c4e3850aa7d4: Pushed
f2a7f0726353: Pushed
latest: digest: sha256:fb9a156ec1b686eed7c055daa08b67f46906dc01ef0ac4d022852ad4da1a5d44 size: 1580
```



- Create a Jenkins pipeline to create a docker image and push the image to Docker hub.

```
pipeline {
    agent any

    stages {
        stage('Build Image') {
            steps {
                sh 'docker build -t myapp .'
            }
        }

        stage('Login to ECR') {
            steps {
                sh '''
                aws ecr get-login-password --region us-east-1 | \
                docker login --username AWS --password-stdin 526018540742.dkr.ecr.us-east-1.amazonaws.com
                '''
            }
        }

        stage('Tag Image') {
            steps {
                sh 'docker tag myapp 526018540742.dkr.ecr.us-east-1.amazonaws.com/myrepo:latest'
            }
        }

        stage('Push Image') {
            steps {
                sh 'docker push 526018540742.dkr.ecr.us-east-1.amazonaws.com/myrepo:latest'
            }
        }
    }
}
```

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```
pipeline {
    agent any

    stages {
        stage('Build Image') {
            steps {
                sh 'docker build -t myapp .'
            }
        }

        stage('Login to Docker Hub') {
            steps {
                sh '''
                docker login -u viqaastaimoor -p dckr_pat_G-mg2GWwI7Oh2Cuj57BJ-KuwFQk
                '''
            }
        }

        stage('Tag Image') {
            steps {
                sh 'docker tag myapp viqaastaimoor/myapp:latest'
            }
        }

        stage('Push Image') {
            steps {
                sh 'docker push viqaastaimoor/myapp:latest'
            }
        }
    }
}
```

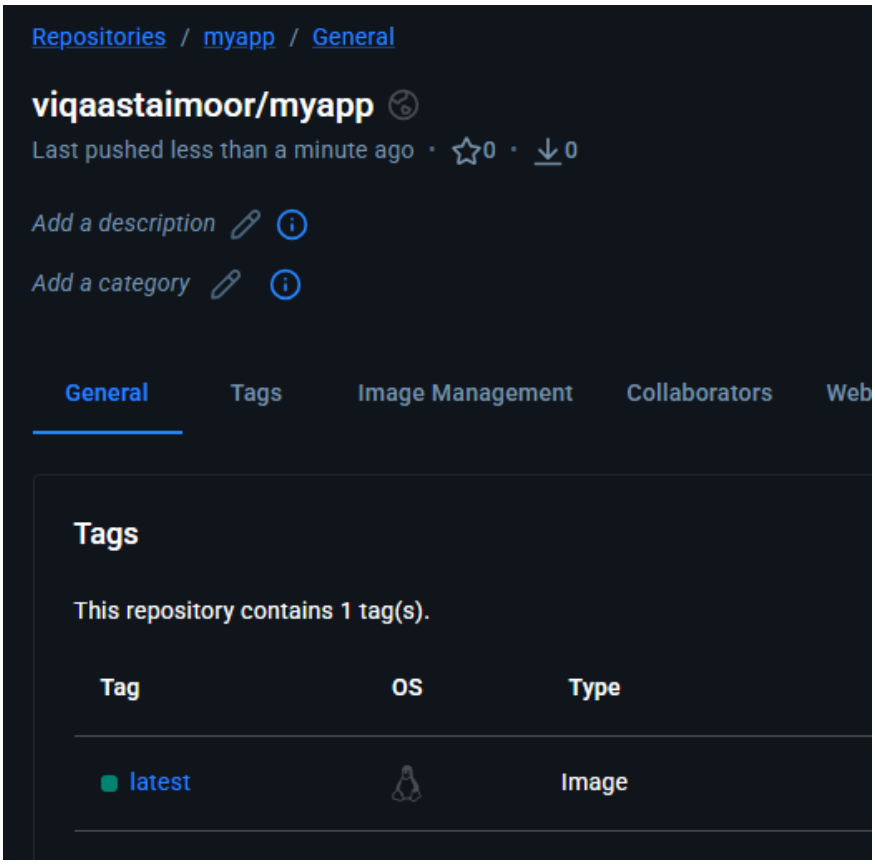
#1 (Mar 26, 2026, 7:29:37 PM)

Started by user admin

This run spent:

- 59 ms waiting;
- 11 sec build duration;
- 11 sec total from scheduled to completion.

No changes.

\

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